TECHNICAL OVERVIEW & RESOURCE GUIDE

Large Language Models & Neural Architectures

An in-depth exploration of Transformer self-attention, autoregressive pre-training, RLHF alignment, and visual learning resources.

TOPIC: Generative AI & Deep Learning **TARGET AUDIENCE:** Engineers & ML Enthusiasts **DATE:** August 2026

1. Executive Summary & Core Mechanics

Large Language Models (LLMs) represent a monumental paradigm shift in artificial intelligence. At their foundation lies the *Transformer architecture*, introduced by Vaswani et al. in 2017. Unlike previous recurrent neural networks (RNNs) or long short-term memory networks (LSTMs) that processed sequentially, Transformers process entire sequences simultaneously through parallelized compute.

$$Attention(Q, K, V) = softmax(\{Q\ K^T\} / \sqrt{d_k})\ V$$

Scaled Dot-Product Attention: Computing context vector representation dynamically.

In this mathematical framework, Q (Query), K (Key), and V (Value) vectors are derived from input token embeddings. The scaling factor $\sqrt{d_k}$ prevents vanishing gradients during softmax normalization when vector dimensions grow large.

▶ FEATURED YOUTUBE VIDEO (7-MIN VISUAL GUIDE)

Large Language Models, Explained Briefly

Created by Grant Sanderson (3Blue1Brown) in consultation with the Computer History Museum. An extraordinary 7-minute visual breakdown illustrating how transformers convert text into geometric vectors and predict subsequent tokens.

Watch Video on YouTube ↗

URL: `https://www.youtube.com/watch?v=WMcwoIyK4DA`

2. Training Pipeline Lifecycle

Phase 1: Unsupervised Pre-training

Models ingest trillions of tokens from public datasets. The objective function minimizes cross-entropy loss by predicting the next token $P(w_t \mid w_1, \ldots, w_{t-1})$, building a massive probabilistic worldview.

Phase 2: Supervised Fine-Tuning (SFT)

Raw base models are fine-tuned on high-quality instruction-response pairs, transforming open-ended text completion into a structured assistant behavior format.

Phase 3: Preference Alignment (RLHF / DPO)

Reinforcement Learning from Human Feedback (RLHF) or Direct Preference Optimization (DPO) aligns model responses with human expectations regarding helpfulness, accuracy, and safety.

Phase 4: Reasoning & Agentic Execution

Modern models leverage chain-of-thought (CoT) prompting, tool-use routing, and external memory stores (RAG) to execute multi-step analytical reasoning tasks.

3. Essential Takeaways & Key Terms

- **Tokens & Embeddings:** Words and sub-words map into high-dimensional vector space where semantic relationships are preserved geometrically.
- **Context Window:** The finite sequence length L that an attention mechanism can evaluate simultaneously during inference.
- **Temperature & Sampling:** Hyperparameters controlling the entropy/randomness of probability distribution selection during text generation.